

Cross-Event Generalization and Few-Shot Fine-Tuning of a Compound-Flood Surrogate Model: A Systematic Measurement with SFINCS

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Abstract

Physics-based models of compound flooding are reliable, but their computational cost becomes prohibitive when many runs are required; the value of a surrogate therefore depends on how much capability it retains outside its training distribution. We use SFINCS to generate 81 compound-forcing scenarios for each of six tropical cyclones affecting Jacksonville and the lower St. Johns River (486 ten-day simulations in total), train a U-Net to predict the hourly water-depth field, and answer three questions with three experiments that share one data split: how accurate the surrogate can be on events it has seen, how much is lost on events it has not, and how much of that loss fine-tuning can recover.

*Same-event single-step RMSE is 0.0178 m, about 5.7 times better than the persistence baseline; cross-event (6-fold leave-one-out) single-step RMSE is 0.0204 m, a 15% degradation, with the degradation factor of all six events falling consistently in 1.13–1.18, and a 22% degradation for rollout prediction. Few-shot fine-tuning closes about 77% of the cross-event gap, and **almost the entire gain comes from the first target run** (6.9% reduction already at $N = 1$, fluctuating within $\pm 3\%$ thereafter); freezing the encoder is as accurate as full fine-tuning; training from scratch on the same N gives an error 2.86 times that of zero-shot; source-domain forgetting is $\leq 10\%$. On the same hardware, running the U-Net over the 168 time steps of one event takes about 1/18 of one SFINCS simulation. All six events share the same catchment, so the conclusions concern generalization across forcings rather than across catchments; rollout prediction still depends on real state inputs.*

1. Introduction

Physics-based hydrodynamic models are reliable for simulating compound flooding, but their computational cost becomes the bottleneck wherever many runs are needed. One

way to reduce the cost of a single run is to simplify the physics [1], yet the cost remains substantial when a study needs hundreds or thousands of members: Lee et al. ran SFINCS ensembles of 81 / 189 / 1000 members [2], the storm-surge surrogate of Kyprioti et al. used 156 synthetic storms [3], and Fraehr et al. treat “how to choose training events under a limited computational budget” as a research question in its own right [4]. The surrogate approach is to use the output of the physics-based model as ground truth for training a data-driven model whose inference is orders of magnitude faster.

Surrogate models and sources of ground truth. Existing work covers a wide range of models and ground truths: a random-forest surrogate of TUFLOW [5], a CNN surrogate of LISFLOOD-FP [6], a U-Net predicting maximum water depth from terrain and rainfall pattern [7], and a transferability study using the maximum water depth of a 2D model as ground truth [8] — all four predict the **event maximum water depth**, whereas this work predicts the hourly water-depth field. Closer to our setting are the autoregressive approaches: a graph network trained single-step with curriculum-style lengthening of the rollout [9], and a neural operator performing long rollouts from an initial field plus time-varying boundaries [10]. The physics-guided LSG hybrid is about 12 times faster than the high-resolution model [11, 12] and has already been combined with SFINCS for compound flooding [13], which is the model combination closest to ours. A very small number of events is the norm in this field: Radfar et al. had only 6 storms, 4 for training, 1 for validation and 1 held out [14].

Cross-event generalization. Several studies have evaluated across events: a 16 / 4 train–test split by whole events [5], 6 for training with 2 held out [15], 10 training catchments evaluated on 5 unseen catchments [16], and a comparison study designed specifically to evaluate events outside the training range [17]; Feng et al. report that a U-

Net degrades markedly on held-out hurricanes [18], a risk we were aware of when choosing a U-Net. These studies differ in model, ground truth, number of events and evaluation protocol, however, so their conclusions are scattered and hard to compare side by side. **What is missing is a systematic comparison that measures, under one and the same setup and on the same set of events, both the “same-event accuracy ceiling” and the “cross-event loss”**: only once we know how well the model can do when it has seen the event can we say how much is lost when it has not. Experiment 1 of this work provides exactly that reference ceiling.

Few-shot fine-tuning. Both the number of target samples N and the conclusions are inconsistent across studies: ≥ 4 labeled samples already suffice [19], an order-of-magnitude reference at $N = 1$ [20], $N = 10$ [21], $N = 30$ [22], while Dheeshjith et al. report that with too few samples fine-tuning is **worse than zero-shot** [23] — which is why the N curve must include $N = 0$. Methodologically, a U-Net transferred with a frozen encoder outperforms training from scratch in the target domain [8], freezing the backbone saves about 50% of training time and 35% of GPU memory [24], and stagewise fine-tuning should lower the learning rate by an order of magnitude [25]; naive fine-tuning may trigger catastrophic forgetting [21]. These conclusions come from different tasks, and whether they carry over to the present setting has to be verified here. Further points of comparison are given in [26–31].

The three experiments address the three questions above in turn: **Experiment 1** (same event) establishes the accuracy ceiling as a reference, **Experiment 2** (cross-event, 6-fold leave-one-out) measures the generalization loss, and **Experiment 3** (few-shot fine-tuning) measures how much of it can be recovered, how the methods differ, and where the marginal return falls off.

Contributions. (1) A controlled cross-event testbed: 486 simulations under one catchment, one grid and one evaluation protocol, so that the “same-event ceiling” and the “cross-event loss” can be subtracted directly on the same test runs. (2) A quantitative measurement of the cross-event loss: 15% single-step degradation with a consistent factor across all six events (1.13–1.18); 22% rollout degradation, which is more sensitive to whether a similar event is present in the training set. (3) An N curve for few-shot fine-tuning and its knee: 270 models show that fine-tuning closes about 77% of the gap, with almost all of the gain in the first target run; freezing the encoder is on a par with full fine-tuning; and training from scratch on the same N cannot replace a pre-trained starting point. (4) An explicit statement of the boundaries of these conclusions: generalization across forcings rather than across catchments, and rollout still depend-

Table 1. Simulation windows of the six events (UTC, all at 00:00)

Hurricane	Sim. start	Event start	Sim. end
Beryl (2012)	2012-05-22	2012-05-25	2012-06-01
Matthew (2016)	2016-10-01	2016-10-04	2016-10-11
Irma (2017)	2017-09-06	2017-09-09	2017-09-16
Dorian (2019)	2019-08-29	2019-09-01	2019-09-08
Ian (2022)	2022-09-23	2022-09-26	2022-10-03
Milton (2024)	2024-10-05	2024-10-08	2024-10-15

The surrogate predicts target times $t = 73 \dots 240$ h.

ing on real state inputs.

2. Study Setup

2.1. Events and Forcing Scenarios

We select six tropical cyclones affecting Jacksonville and the lower St. Johns River according to two criteria: **data completeness** (all three forcings — seaward water level, river discharge and rainfall — covering the simulation window in full) and **severity of impact on the study area** (measured by the peak non-tidal residual the event produced at the Mayport station). The simulation window and time conventions are given in Section 2.2; the surrogate predicts target times $t = 73, \dots, 240$ h within the event segment (Table 1). The rainfall product differs by data availability: Irma, Matthew and Dorian use MRMS GaugeCorr_QPE_01H, Ian and Milton use MRMS Multi-Sensor_QPE_01H_Pass2, and **Beryl (2012) predates the MRMS record, so NCEP Stage IV 01h is used instead (4 km native grid, coarser than the 1 km of MRMS)**.

For each event, 81 compound-forcing scenarios are generated on top of the observed forcings as the full factorial of four factors at three levels each ($3^4 = 81$, Table 2):

$$h_s = h_{\text{tide}} + \alpha (h_{\text{obs}} - h_{\text{tide}}), \quad (1)$$

$$P_s = \beta P(x, t - \tau), \quad (2)$$

$$Q_s = \gamma Q_{\text{low}} + (Q - Q_{\text{low}}). \quad (3)$$

All quantities are functions of t ; h_{tide} is the Mayport astronomical tide prediction and Q_{low} is a 25 h centered moving average. Scenarios are numbered s001–s081 in lexicographic order of $(\alpha, \beta, \gamma, \tau)$, with s041 as the baseline. The six events give **486 SFINCS simulations** in total.

Scope of the scenarios. These are “compound-forcing scenarios”, not “hurricanes whose category has been changed”: the perturbations act only on the three forcing time series. The model takes no wind or pressure field as input, and the observed Mayport water level already contains the surge, so **what α scales is “the surge amplitude this storm produced at Mayport”, not “moving the hurricane onto a different track”**; nor is the non-tidal residual

Table 2. The four perturbation factors of the compound-forcing scenarios

Factor	Values	Applied to
Rainfall scale β	0.70 / 1.00 / 1.30	Per-cell rainfall intensity over the whole domain
Non-tidal residual scale α	0.80 / 1.00 / 1.20	Non-tidal residual of the Mayport water level
Low-frequency discharge scale γ	0.75 / 1.00 / 1.25	Low-frequency component of the Acosta discharge
Rainfall time shift τ (h)	-6 / 0 / +6	Time phase of the rainfall field

The full factorial of the four factors gives 81 scenarios; the baseline scenario s041 is $\alpha = \beta = \gamma = 1$, $\tau = 0$.

the same thing as pure storm surge. The values of the four factors constitute the sensitivity-experiment design of this work and do not amount to a probabilistic characterization of real climate scenarios.

2.2. SFINCS Model

The simulation window of each event is **10 days = 72 h spin-up + 168 h event segment**, all in UTC; the hour label T denotes the interval $[T - 1 \text{ h}, T)$, giving 240 labels per event. The study area is centered on two HUC10 subbasins of the USGS Watershed Boundary Dataset: **0308010315 “Trout River–Saint Johns River” (360.07 km²)** and **0308010316 “Saint Johns River–Atlantic Ocean” (614.4 km²)**, with the easternmost boundary extended somewhat seaward so that the water-level boundary can be imposed, giving an active model domain of 977.49 km² (Figure 1). The terrain merges the USGS 3DEP 1 arc-second DEM with the NOAA CRM (taking the smaller value only where the DEM is ≤ 0.5 m and keeping the DEM elsewhere), re-sampled to 60 m and then block-averaged onto the computational grid.

The computational grid is **195 rows (south→north) $\times$ 255 columns (west→east)** at 200 m spacing in EPSG:26917 (SFINCS $n_{\text{max}} = 195$, $m_{\text{max}} = 255$; field variables are indexed consistently as (row, column)). There are 24,467 active cells, of which 95 are water-level boundary cells, and 21,933 land cells (active with $z_b \geq 0$). On the seaward side, 12 boundary points share a single observed total water-level series from Mayport (8720218); upstream, 4 source cells split the signed total discharge from Acosta (02246500), without tidal filtering, by cross-section width (0.32 / 0.15 / 0.38 / 0.15, summing to 1.00); the remaining water cells on the domain boundary are treated as walls. Rainfall is resampled to a 1 km grid and imposed hourly. The main parameters are `manning_land` = 0.06, `manning_sea` = 0.02, `huthresh` = 0.05 m, `advection` = 1 and `qinf` = 0 (infiltration not considered); the initial water level is taken per scenario

as that scenario’s value at `tstart`. Simulations use the Docker image `deltares/sfincs-cpu` with SFINCS v2.4.2-alpha; all 486 runs passed verification.

2.3. Surrogate Model and Experimental Setup

Task. Given the state and forcing before t , predict the water-depth field $h(t)$ at that time (195×255 , m), taking h on wet cells and 0 on dry cells, with inactive cells excluded from the loss. Both evaluation modes are reported: in **single-step prediction** the look-back inputs $h(t-1 \dots t-3)$ are all SFINCS ground truth; in **rollout prediction** the model starts from the ground-truth $h(70 \dots 72)$ and, from $t \geq 73$ onwards, feeds its own predictions back in and advances 168 steps in succession (rainfall, water level and discharge always use ground truth). **This work is primarily concerned with single-step prediction**; rollout is reported as usual but is not an optimization target.

Network. U-Net: 4 downsampling stages, encoder $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$, bottleneck 512, skip connections concatenated along channels, output 1×1 convolution + ReLU, 7,765,921 parameters; the architecture is sketched in Appendix Figure 12. There are 13 input channels: bed elevation 1, water-depth look-back 3, rainfall look-back 3, seaward water level 3, river discharge 3 (the last two are scalars, each broadcast over the whole map as a constant channel). The 195×255 field is padded to 208×256 before the network and the output is cropped back; both the padded cells and the inactive cells are excluded from the loss. Layer-by-layer channels are listed in Appendix Table 9.

Training. Adam with a fixed learning rate of 10^{-3} ; MSE masked to active cells; batch size 8; at most 50 epochs, with early stopping when the validation loss has not decreased for 10 epochs; training and validation take a fixed set of 24 time steps per run (one every 7 h), while testing evaluates all 168 time steps without subsampling; inputs are z-scored per channel group, with the statistics computed only from the respective training set. The full hyper-parameters are given in Appendix Table 10.

Splits. The 81 scenarios of each event are split once into 60 training / 11 validation / 10 test runs, shared by all three experiments. **Experiment 1** trains a single model (the *all-event model*) on the six events combined (360 train / 66 val) and reports separately on the 10 test runs of each event. **Experiment 2** trains 6 *leave-one-out models*, each fold with 300 train / 55 val / 10 test. **Experiment 3** starts from the best model of each Experiment 2 fold and fine-tunes on $N = 1/2/3/5/10$ runs drawn from the training pool of the held-out event ($N = 0$ is zero-shot and is taken directly from the Experiment 2 result), using three methods:

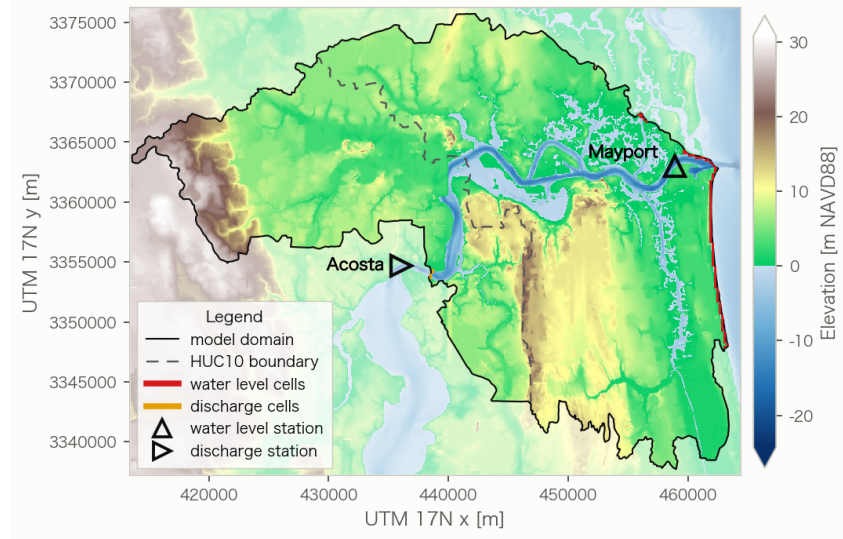


Figure 1. Study area and SFINCS model setup. The base map is the 60 m merged terrain (m NAVD88), lightened outside the model domain; the grey dashed line is the HUC10 boundary between the two subbasins (identifiers and names in the text); $\triangle$ is the Mayport 8720218 water-level station and $\diamond$ the Acosta 02246500 discharge station.

full fine-tuning (10^{-4} , 20 epochs); **frozen-encoder fine-tuning**, freezing the encoder blocks and the bottleneck and tuning only the decoder side (10^{-4} , 20 epochs); and **training from scratch**, random initialization trained on those N runs only (10^{-3} , 50 epochs). None of the three uses early stopping or a validation set, so that training from scratch does not gain extra exposure to target-event data. The N runs are drawn stratified by forcing level and nested, and the three methods use the same set of runs for the same fold, N and seed; each configuration uses 3 random seeds, giving 270 models in total. Each fine-tuned model is additionally evaluated single-step on the 50 test runs of the other five events to measure source-domain forgetting.

Baseline and metrics. The persistence baseline is $h(t) = h_{\text{true}}(t - 1)$. The metrics are MAE, RMSE (m) and $\text{CSI} = \text{TP}/(\text{TP} + \text{FP} + \text{FN})$ (wet = $h > 0.1$ m or 0.3 m), each reported on active cells and on land cells. For each event the metrics are computed over all of its test runs, time steps and cells together; the value reported for all six events is the arithmetic mean of the six per-event values, so that every event carries equal weight.

Inference speed. On the same Mac, the median over the 486 measured runs for SFINCS to complete one 10-day simulation is **35.2 s**; the U-Net takes **1.92 s** in total to run the same event over its 168 event time steps (11.4 ms per time step, MPS, batch 8, forward pass only), a factor of about **18**. On the A100 a single time step takes about 4.2 ms, but that is different hardware from the SFINCS timing and the two

cannot be divided directly.

3. Results

3.1. Same-Event and Cross-Event

Single-step prediction is clearly better than the persistence baseline, and the cross-event degradation is limited and highly consistent across the six events (Table 3). Averaged over the six events, the same-event single-step RMSE is 0.0178 m against 0.1017 m for persistence, a factor of about 5.7; all 60 test runs beat the baseline, and so does every single time step of every event. Averaged over the six cross-event folds the RMSE is 0.0204 m, 15% higher than same-event, and 60/60 still beat the baseline; the per-event degradation factors are 1.13 for Irma, 1.18 for Matthew, 1.13 for Ian, 1.14 for Milton, 1.17 for Dorian and 1.13 for Beryl, so **all six fall in 1.13–1.18**. The single-step CSI (threshold 0.1 m) is 0.976 same-event and 0.974 cross-event, almost identical. Water depth changes little within an hour, so persistence is not a weak baseline; beating it by more than a factor of 5 is what shows that the model has learned how the depth changes.

Rollout prediction, by contrast, accumulates error over time, degrades more across events, and varies more between events. The six-event mean rollout RMSE is 0.2412 m same-event and 0.2931 m cross-event (a factor of 1.22), both higher than persistence; the per-event factor ranges from 0.74 for Matthew (which is in fact better) to 2.05 for Dorian, far less consistent than the single-step case.

The error grows with the distance from the last real input and jumps during the most intense part of the

Table 3. Main metrics for Exp. 1 (same event) and Exp. 2 (cross-event, 6-fold leave-one-out). RMSE in m, all on active cells

Event	Single-step RMSE		Baseline RMSE	Rollout RMSE	
	Exp. 1	Exp. 2		Exp. 1	Exp. 2
Irma	0.0214	0.0242	0.0963	0.3697	0.4395
Matthew	0.0185	0.0218	0.0926	0.2696	0.1999
Ian	0.0185	0.0209	0.1065	0.1613	0.2233
Milton	0.0156	0.0178	0.0946	0.2835	0.3149
Dorian	0.0149	0.0174	0.1182	0.1421	0.2916
Beryl	0.0182	0.0205	0.1022	0.2211	0.2893
Mean	0.0178	0.0204	0.1017	0.2412	0.2931

Baseline = persistence, i.e. the SFINCS truth at $t - 1$ used as the prediction for t ; it is model-independent and shared by both experiments. Each Exp. 2 row is the result obtained when that event is the held-out event. Each event row pools the 10 test runs of that event $\times$ 168 time steps; the Mean row is the arithmetic mean of the six event rows.

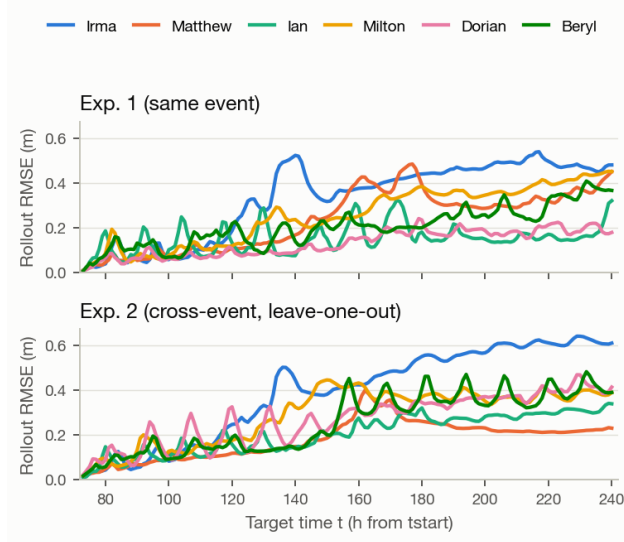


Figure 2. Rollout error over time (active-cell RMSE, the 10 test runs of each event pooled). Top: Experiment 1 (same event); bottom: Experiment 2 (cross-event, leave-one-out). Started from the ground-truth $h(70-72)$.

flood (Figure 2). Irma jumps abruptly near $t \approx 130$ h, which corresponds to the s041 flood peak at $t = 129$ h. The time at which the rollout RMSE first exceeds persistence at the same time step is 6–70 h after the start for same-event, shortening to 7–26 h for cross-event.

Spatially, having seen the event or not changes the magnitude of the error, not its structure (Figure 3). Experiment 1 has the smallest error for all three events and Experiment 2 is higher throughout, but in both cases the errors are fine-grained and alternate in sign, with no systematic bias. This is one time step of one run shown as an example; the quantitative conclusions rest on Table 3, and the complete metrics including land cells, MAE and CSI 0.3 are given in Appendix Tables 7 and 8.

3.2. Few-Shot Fine-Tuning

The pre-trained starting point is decisive for few-shot fine-tuning (Figures 4 and 5). Even with $N = 10$ runs, training from scratch still gives a six-fold mean single-step RMSE of 0.0584 m, **2.86 times that of zero-shot (0.0204 m)**; its error does fall with N (from 0.1092 at $N = 1$ to 0.0584 at $N = 10$), showing that it is indeed fitting the target event, but it comes nowhere near the pre-trained model. Both fine-tuning methods are on average better than zero-shot at every N .

Fine-tuning works, but its magnitude is limited, and almost all of the gain comes from the first target run. The six-fold mean single-step RMSE falls from 0.0204 to 0.0184 (full fine-tuning, $N = 5$), a 9.8% reduction; same-event Experiment 1 is 0.0178 m and the cross-event degradation itself is only 13–18%, so fine-tuning closes about **77% of it (0.0020/0.0026)**. The reduction relative to zero-shot is 6.9% at $N = 1$, 8.3% at $N = 2$, 7.1% at $N = 3$, 9.8% at $N = 5$ and 9.0% at $N = 10$ — **the fluctuation beyond $N \geq 2$ stays within $\pm 3\%$, the same order as the 3-seed range, and no second knee is visible.**

Freezing the encoder is as accurate as full fine-tuning; the clear difference in stability is between fine-tuning and training from scratch. At $N = 10$, full fine-tuning gives 0.0186 and frozen-encoder fine-tuning 0.0185, and the relative difference per fold and per N swings between -4.1% and $+10.9\%$ with no consistent direction (the points in Appendix Figure 9 lie essentially on the 1:1 line). The 3-seed range is narrow for full fine-tuning and wide for training from scratch, which indicates that fine-tuning is stable whereas training from scratch is not; the per-fold curves are given in Appendix Figure 10. Active cells and land cells lead to the same conclusion.

Every event benefits and the direction is consistent, but the magnitude shows no visible relation to event attributes (Figure 6). The magnitude is 6.4–15.7% (Milton 6.4, Matthew 7.5, Ian 7.9, Irma 8.4, Dorian 13.0, Beryl 15.7%), with Dorian and Beryl about twice the other four; after fine-tuning, Beryl and Dorian have reached or surpassed Experiment 1 (Beryl $0.0172 < 0.0181$, Dorian $0.0151 \approx 0.0148$), and the Beryl fold already reaches a 3-seed mean of 0.0175 m at $N = 1$. The size of the improvement shows no visible relation to wetness, total rainfall or the Experiment 2 degradation factor (Matthew degrades the most and improves the least, Beryl degrades the least and improves the most); with only 6 events this is a qualitative observation only (Appendix Table 6).

Spatially, the error structure of both fine-tuning methods is of the same kind as that of the pre-trained model, whereas training from scratch shows a systematic bias (returning to Figure 3). After fine-tuning, the errors of both fine-tuning methods are mostly lower than zero-shot but still higher than Experiment 1, the excep-

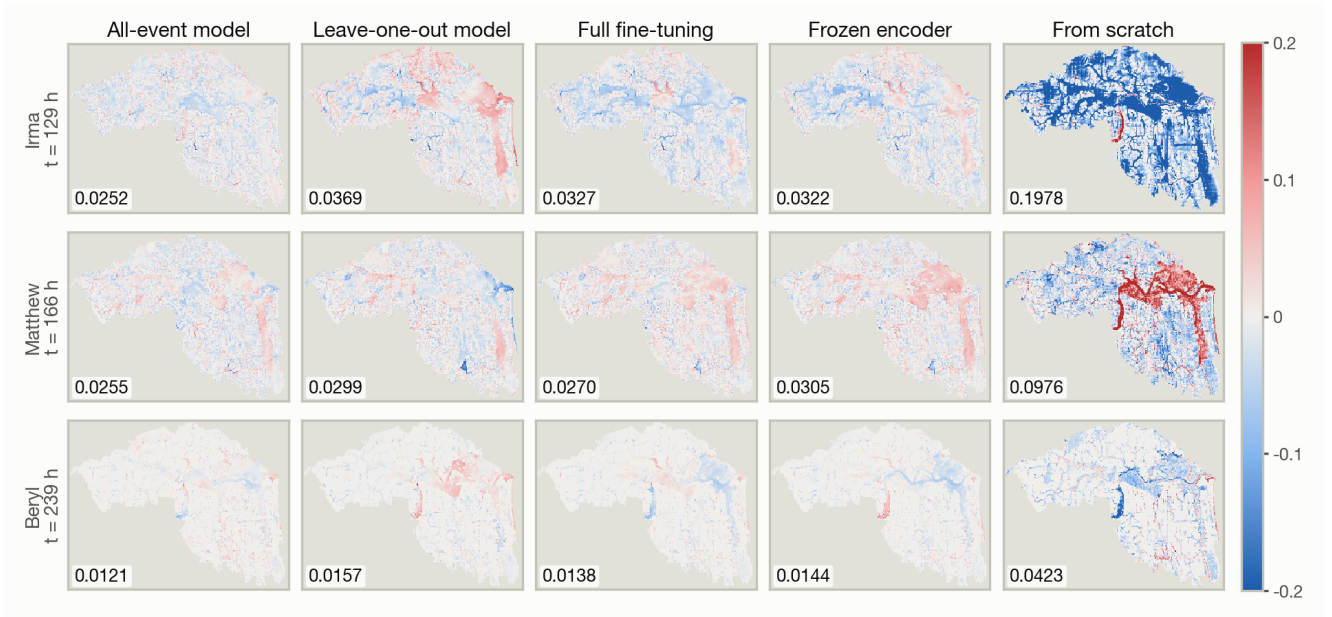


Figure 3. Single-step prediction error for Irma / Matthew / Beryl at their respective s041 flood peaks (= prediction – ground truth, m). The columns are, in order, Experiment 1, Experiment 2 (zero-shot), and Experiment 3 full fine-tuning, frozen-encoder fine-tuning and training from scratch, the last three all using the $N = 10$, seed 0 weights of that fold; the label in the lower left corner is the active-cell RMSE (m) of that panel.

tion being frozen-encoder fine-tuning for Matthew (0.0305 m), slightly worse than zero-shot (0.0299 m); training from scratch shows large areas of same-signed bias (underestimation over the whole domain for Irma; overestimation in the main channel and underestimation elsewhere for Matthew), with an RMSE 2.7–7.9 times that of the other models.

Rollout prediction likewise flattens out after $N = 1$, while training from scratch frequently diverges. Both fine-tuning methods improve the rollout by about 12–18% (from 0.2931 zero-shot to 0.2505–0.2583 for full fine-tuning and 0.2391–0.2568 for frozen-encoder fine-tuning). There are **53 configurations with a rollout RMSE > 0.6 m, all of them from training from scratch** (the maximum is 0.3817 for full fine-tuning and 0.3559 for frozen-encoder fine-tuning), the most extreme being Matthew at $N = 3$, seed 1, at 44.6 m. **Any statement about the rollout of training from scratch uses the median and states the number of outliers; the mean is never quoted:** the median over its 18 configurations at each N is 0.750 / 0.690 / 0.632 / 0.604 / 0.614 m (with 5 / 0 / 3 / 5 / 5 of them above 1 m). The single-step error of these configurations is not anomalous (0.05–0.17 m), which indicates that the divergence comes from autoregressive amplification rather than from a collapse of the single-step prediction.

The gain from fine-tuning does not favor any particular class of scenario, and its cost is a slight source-domain forgetting. Stratified by forcing level and by rainfall time shift, the reduction from zero-shot to full fine-

tuning at $N = 5$ is 9.5–10.1% throughout (Appendix Table 4); after fine-tuning, the single-step RMSE on the 50 test runs of the other five events rises from 0.0184 to 0.0194 ($N = 1$) and 0.0203 m ($N = 10$), a degradation of 5.6% to 10.3%, at most +17.3% for a single fold, and slightly less for frozen-encoder fine-tuning (4.0% to 8.1%) (Appendix Table 5). The magnitude is far smaller than the gain on the target event, and the result stays an order of magnitude better than persistence throughout.

4. Discussion

4.1. Same-Event and Cross-Event

What the current model depends on is a continuous supply of real state inputs, not more training data. The rollout error accumulates monotonically with time and jumps during the flood peak, which indicates that the deficiency lies in having no real state available at inference time; more precisely, the model **needs a continuous supply of real state inputs (observation assimilation, for instance)** rather than “more ground-truth training data”. The spatial distribution of the rollout error is too high in some areas and too low in others, with no single direction, which may serve as a lead for improving the rollout.

The comparison with persistence must distinguish between protocols. Persistence is comparable with single-step prediction under the same protocol — both see the ground-truth $h(t - 1)$ at every step; it is under a differ-

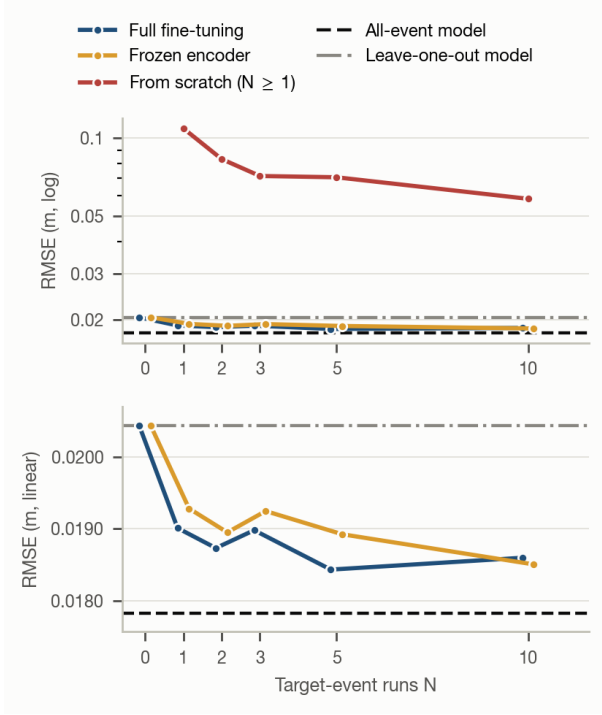


Figure 4. **Active cells**: single-step RMSE as a function of the number of target-event runs N (six-fold mean, each fold being a 3-seed mean). Top: overview including training from scratch (log scale); bottom: zoom without it (linear scale). The black dashed line is the six-fold arithmetic mean of the Experiment 1 same-event models (0.0178 m) and the grey dash-dotted line is the six-fold mean of the Experiment 2 zero-shot models; training from scratch is drawn from $N = 1$ onwards (it does not exist at $N = 0$). Land cells are shown in Figure 5.

ent protocol from rollout, which sees ground truth only at the starting point. That rollout is worse than persistence is a statement about error accumulation and must not be read as “the model is worse than copying”.

The CSI on active cells is inflated by cells that are permanently wet. The active-cell CSI differs little between the three modes (0.976 single-step, 0.758 rollout, 0.947 persistence), possibly because the active cells include channel cells and the like that hold water all year round and are almost always “guessed right”; on land cells the differences are much more pronounced (0.951 / 0.555 / 0.898).

The fluctuation of the validation loss across the Experiment 2 folds cannot be attributed (Appendix Figure 7). Each fold is trained with a single random seed, so training randomness is not controlled; the Irma fold is comparatively smooth while the Dorian and Beryl folds fluctuate more, which may come either from the validation set containing extreme events or from the randomness of training itself. This point is a conjecture only. Note that the validation set of each fold is not the held-out event but the

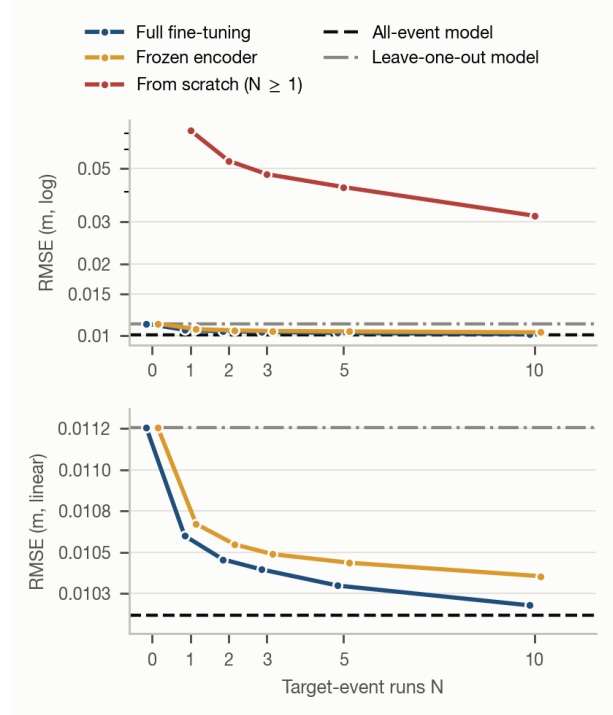


Figure 5. **Land cells** (active with $z_b \geq 0$): all other conventions and reference lines as in Figure 4. The conclusions agree under the two masks.

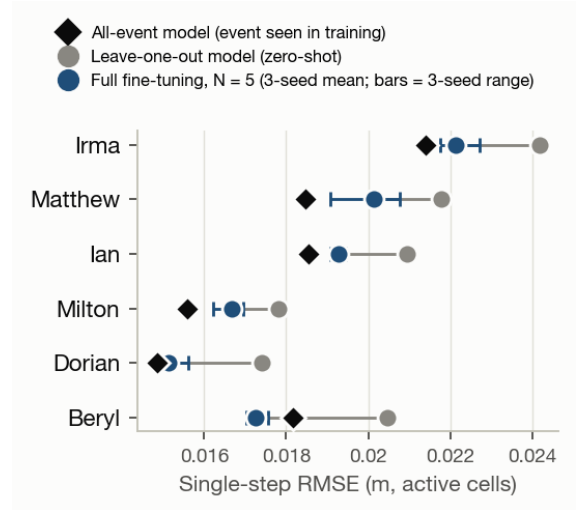


Figure 6. Experiment 3: per-event single-step error before and after fine-tuning compared with Experiment 1 (active-cell RMSE), the 10 test runs of each event pooled. Full fine-tuning at $N = 5$ is a 3-seed mean and the bars are the 3-seed range.

55 validation runs of the other five events.

4.2. Few-Shot Fine-Tuning

The pre-trained starting point is decisive, and the immediate reason is the difference in the amount of data. The pre-trained starting point used $5 \text{ events} \times 60 = 300$ runs, whereas training from scratch has only $N \leq 10$. Another conjecture: each event was perturbed into 81 scenarios, which may have let the pre-trained model learn a richer set of regularities in how the water depth changes; this experiment does not test that separately, and “a control experiment that pre-trains on the original scenarios only” is listed as future work.

Frozen encoder $\approx$ full fine-tuning: we reproduce “better than training from scratch” but not “freezing beats full fine-tuning”. The direction agrees with Selem et al. [8] (both fine-tuning methods are far better than training from scratch), but the two are equally accurate. Lehmann et al. [24] report that freezing the backbone saves about 50% of training time and 35% of GPU memory; this experiment did not measure resource consumption, so we can only say that the accuracy is on a par.

“Fine-tuning worse than zero-shot” did not occur. This differs from what Dheeshjith et al. [23] report (referring to the 3-seed mean; for individual seeds, such as Irma at $N = 10$, the range does overlap with zero-shot). There are four possible reasons, all of which are conjectures: (1) the gap between the source and target domains is small — the same catchment and the same grid, with only the forcing differing, and zero-shot is only 13–18% worse than same-event; (2) each sample is rich in information — $N = 1$ already means 24 time steps, each sample is a full water-depth map and about 24 thousand active cells all provide a supervision signal, so the N of different studies cannot be compared as bare numbers; (3) the fine-tuning setup is conservative — a learning rate lower by an order of magnitude and a fixed 20 epochs, whereas the pilot run showed the error rebounding towards the end of 60 epochs, so 20 epochs avoids overfitting; (4) the single-step task is comparatively local — the input contains the real water depth of the previous 3 hours and the model mainly learns “how it changes over the next hour”, which a small amount of target data is unlikely to throw off.

The largest gain, on Beryl, may be related to the rainfall product. Beryl, which improves the most, is the only event using Stage IV, and it also has the largest bias in maximum inundation area in Experiment 2 (−46.8%). One possible explanation is that the systematic bias introduced by the difference in forcing product happens to be exactly the part that a small amount of target data corrects most easily. This is a single-point observation, not a conclusion.

4.3. Why the Cross-Event Degradation Is Small: An Observation That Cannot Be Separated

The seaward water level and the river discharge of the six events are fairly close in trend and magnitude, but **the rainfall differs greatly — the domain-mean accumulation over the event segment ranges from about 42 mm (Ian) to about 304 mm (Irma)** (Appendix Figure 8). That the water level and discharge are close may be one reason why the cross-event single-step degradation is small; but the input of the single-step prediction itself contains the ground-truth look-back water depth of the previous 3 hours, and that factor would depress the cross-event difference just as well. **The two factors cannot be separated within this experimental design;** this is put forward here as an observation, not a conclusion.

5. Limitations

1. *Model setup:* infiltration is not considered ($q_{\text{inf}} = 0$); the 200 m grid does not represent the four tributary creeks (they are already absent from the 30 m source DEM), and the corresponding stations are not quantitatively validated.
2. *Cross-event means across forcings, not across catchments:* the six events share one terrain and one grid, so the model is never required to generalize to new terrain; this work cannot answer how well it would transfer across catchments.
3. *There are only 6 effectively independent events:* Experiment 2 uses one random seed per fold and Experiment 3 uses 3 seeds per configuration, so the differences between folds contain both event difficulty and training randomness.
4. *The training budget did not reach convergence:* 5 of the 6 Experiment 2 folds ran the full 50 epochs with their best epoch close to the limit; training from scratch has not converged at 50 epochs (in the pilot run, $50 \rightarrow 80$ epochs gave $\text{RMSE } 0.049 \rightarrow 0.044$), but even extended to 80 epochs it remains more than an order of magnitude worse than full fine-tuning, so this does not affect the direction of the conclusion. The setup was left unchanged to keep the folds comparable.
5. *The rainfall product is not uniform:* Beryl uses Stage IV at 4 km and the rest use MRMS at 1 km (and MRMS itself is two product generations, before and after 2021).
6. *Experiment 3 uses a fixed number of epochs, no early stopping and no validation set:* all of its numbers are results under “a fixed 20 / 50 epochs”, not the best result of each configuration.
7. *The rollout prediction uses ground-truth forcing,* so it does not include forecast error in the forcing; the data are hourly snapshots, and what happens between hours is outside the scope of the evaluation.

8. *The improvement is small in absolute terms*: the cross-event degradation is itself only 13–18%, which limits the room fine-tuning has to improve on; the conclusions must not be extrapolated to cross-catchment settings or to settings where the forcing differs more.

6. Conclusion

This work builds a controlled cross-event testbed for Jacksonville and the lower St. Johns River — 81 compound-forcing scenarios for each of six tropical cyclones, 486 SFINCS simulations as ground truth — trains a U-Net to predict the hourly water-depth field, and uses three experiments sharing one data split to measure, in turn, the same-event accuracy ceiling, the cross-event generalization loss, and how much of that loss few-shot fine-tuning can recover.

Same-event single-step RMSE is 0.0178 m, about 5.7 times better than persistence; the cross-event degradation is 15% with a consistent factor across the six events (1.13–1.18), while the rollout degradation is 22% and is more sensitive to whether a similar event is present in the training set. Few-shot fine-tuning closes about 77% of the cross-event gap, with almost all of the gain in the first target run; freezing the encoder is on a par with full fine-tuning; training from scratch on the same N cannot replace a pre-trained starting point; and source-domain forgetting is $\leq 10\%$. On the same hardware, running the U-Net over the 168 time steps of one event takes about 1/18 of SFINCS.

Three directions for future work follow from the observations made here: (1) **reducing the dependence of rollout prediction on real state inputs** (multi-step training, observation assimilation); (2) **a control experiment that pre-trains on the original scenarios only**, to test what the scenario perturbations contribute to the quality of pre-training; and (3) **reducing the number of training hurricane events**, to see how the cross-event prediction capability varies with the number of training events.

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A. Appendix: Supplementary Figures and Tables

A.1. Supplementary Figures

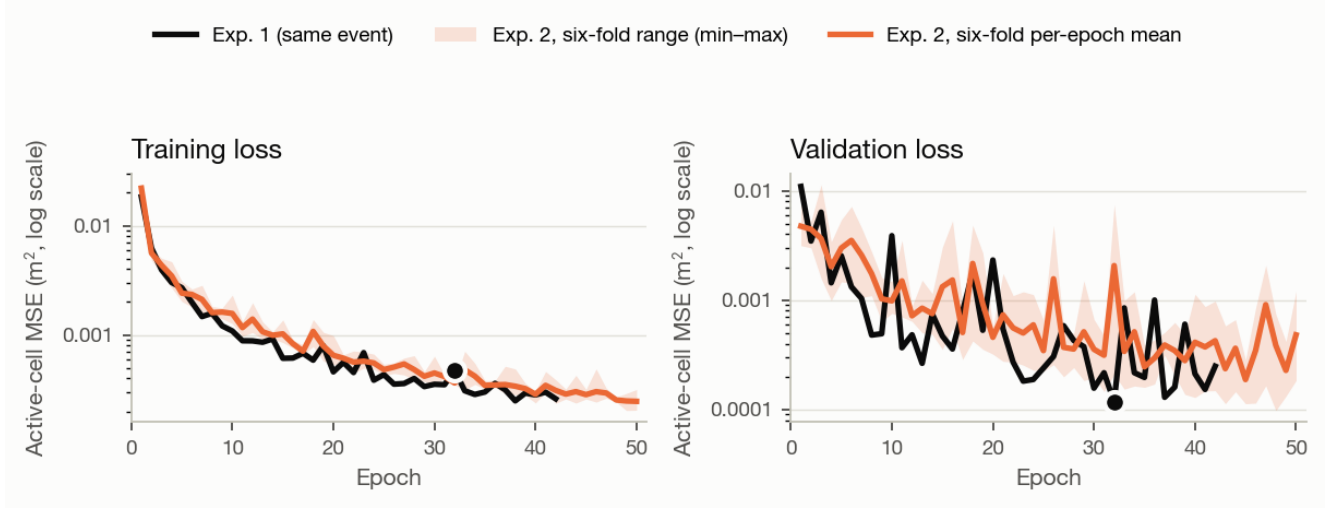


Figure 7. Training curves: Experiment 1 (the black dot is the best epoch selected by validation loss, epoch 32) and the per-epoch mean over the six Experiment 2 folds. The vertical axis is the active-cell MSE (in m^2); at most 50 epochs, with early stopping when the validation loss has not decreased for 10 epochs. The best epoch of the six folds falls between 31 and 48; the Milton fold stopped early at epoch 41, so the mean and range over epochs 42–50 use only the 5 folds that still have data at those epochs.

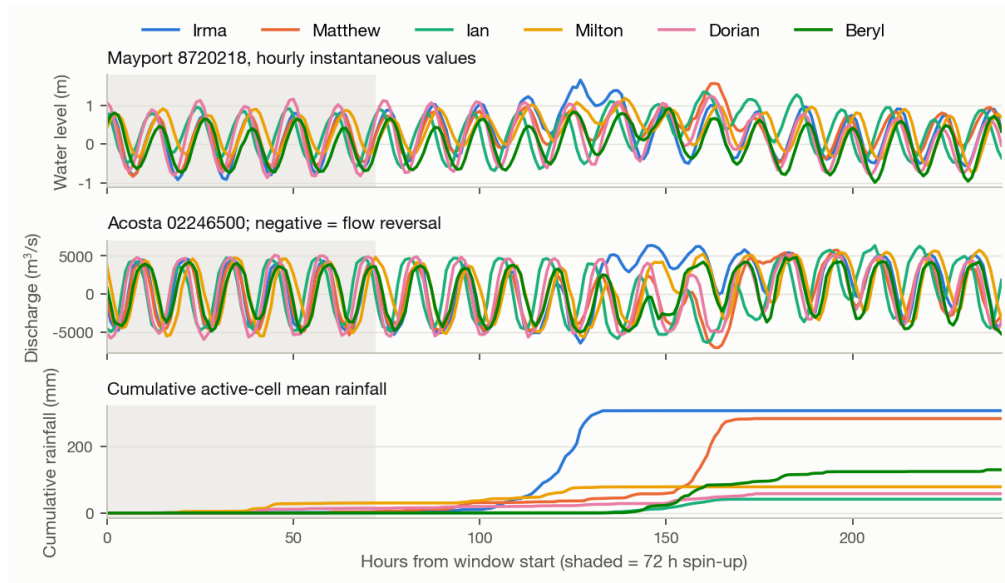


Figure 8. The three forcing time series of the baseline scenario (s041) of the six events, overlaid. The water level is the hourly instantaneous value at Mayport 8720218; the discharge is Acosta 02246500, where negative means flow reversal (upstream flow on the flood tide); the rainfall is the hourly accumulation of the active-cell mean rainfall intensity. The horizontal axis is hours from the start of the window, and the shaded band is the 72 h spin-up segment.

A.2. Supplementary Tables

A.3. U-Net Architecture and Hyper-Parameters

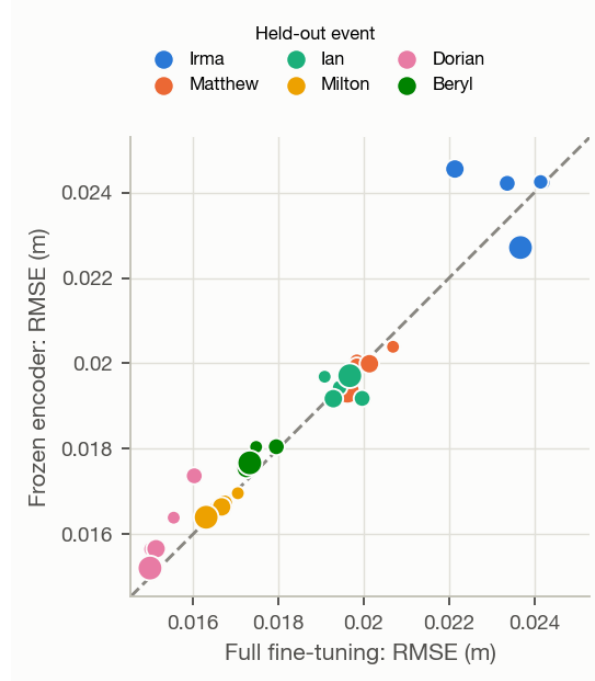


Figure 9. Experiment 3: single-step RMSE of full fine-tuning against frozen-encoder fine-tuning (active cells). Each point is one fold $\times$ one N as a 3-seed mean, larger points meaning larger N ; the dashed line is 1:1.

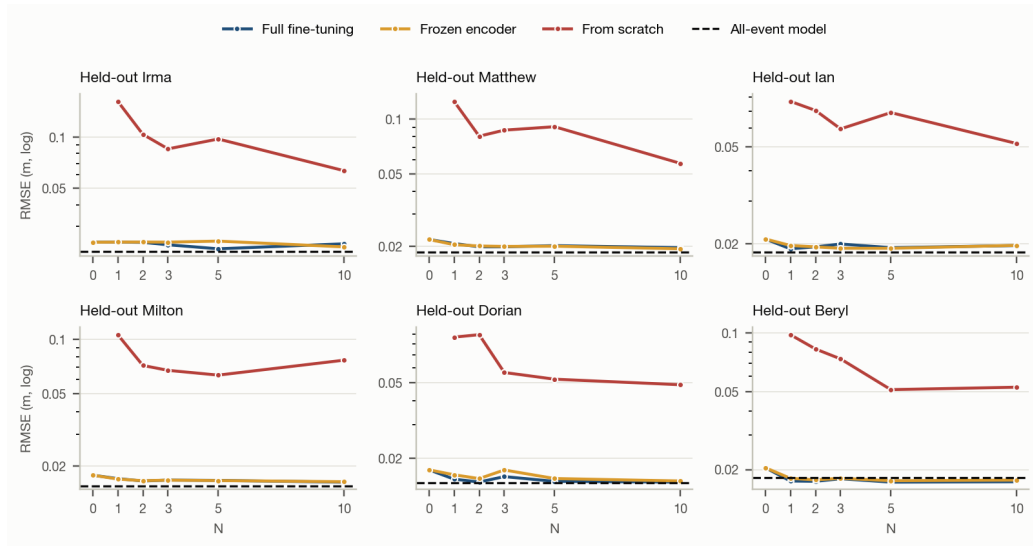


Figure 10. Per-fold N curves of Experiment 3, **active cells** (single-step RMSE, 3-seed mean; the black dashed line is the Experiment 1 all-event model for that event). Land cells are shown in Figure 11.

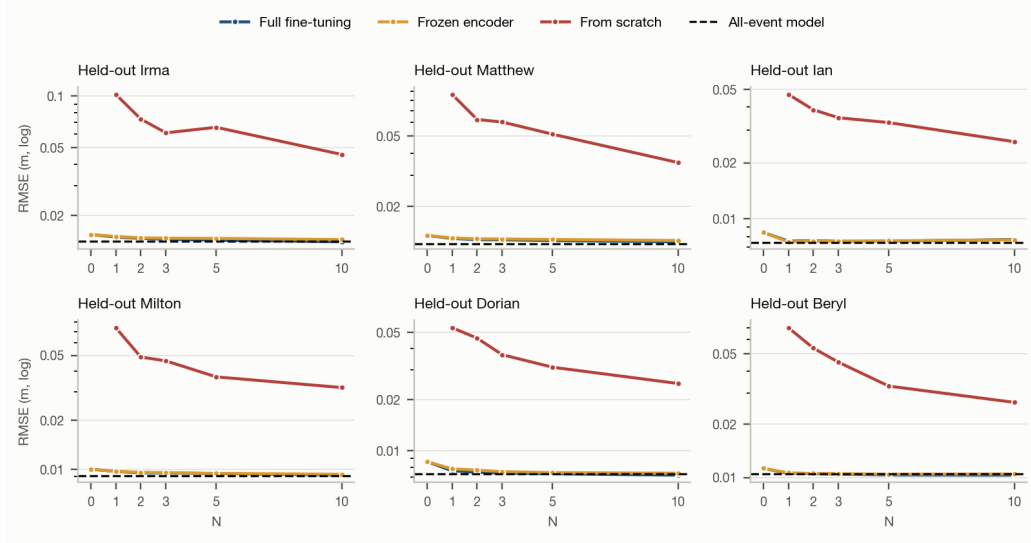


Figure 11. Per-fold N curves of Experiment 3, **land cells**; conventions as in Figure 10.

Table 4. Exp. 3 stratified results: single-step RMSE (active cells, six-fold mean) and the reduction relative to zero-shot

Stratum	Zero-shot RMSE	Full FT, N=5		Full FT, N=10	
		RMSE	Reduction	RMSE	Reduction
Compound forcing level: low	0.0198	0.0179	9.9%	0.0181	8.9%
Compound forcing level: medium	0.0205	0.0185	9.7%	0.0187	9.0%
Compound forcing level: high	0.0209	0.0188	9.7%	0.0190	8.8%
Rainfall time shift $\tau = -6$ h	0.0206	0.0186	9.6%	0.0187	9.0%
Rainfall time shift $\tau = 0$	0.0203	0.0182	10.1%	0.0184	9.0%
Rainfall time shift $\tau = +6$ h	0.0205	0.0186	9.5%	0.0187	8.7%

Full FT = full fine-tuning. The compound forcing level is defined by (α, β, γ) ; τ is the rainfall time shift. 27 scenarios each. Compound forcing levels are scenarios binned by the combined severity of the four perturbation factors.

Table 5. Exp. 3 source-domain forgetting: single-step RMSE on the 50 test runs of the other five events after fine-tuning (active cells, six-fold mean)

N	Full fine-tuning			Frozen-encoder fine-tuning		
	RMSE	vs. before	Worst fold	RMSE	vs. before	Worst fold
0 (before fine-tuning)	0.0184	—	—	0.0184	—	—
1	0.0194	+5.6%	+10.9%	0.0191	+4.0%	+10.3%
2	0.0195	+6.1%	+12.7%	0.0192	+4.6%	+11.4%
3	0.0200	+8.8%	+16.7%	0.0199	+7.9%	+15.1%
5	0.0198	+7.5%	+15.4%	0.0198	+7.8%	+14.4%
10	0.0203	+10.3%	+17.3%	0.0199	+8.1%	+16.5%

Source domain = the other five events used to train that fold, 10 test runs each; before fine-tuning = the Exp. 2 model of that fold.

Table 6. Exp. 3: per-event attributes and fine-tuning gain (single-step RMSE, active cells)

Event	Full FT, N=5 RMSE (m)	Reduction	Degrad. factor	Wet area (km ²)	Total rainfall (mm)	Rainfall product
Irma	0.0221	8.4%	1.13	479.8	322.1	MRMS
Matthew	0.0201	7.5%	1.18	448.2	259.6	MRMS
Ian	0.0193	7.9%	1.13	94.2	39.3	MRMS
Milton	0.0167	6.4%	1.14	166.4	84.0	MRMS
Dorian	0.0151	13.0%	1.17	107.9	58.8	MRMS
Beryl	0.0172	15.7%	1.13	246.8	120.8	StageIV

Full FT = full fine-tuning. The Exp. 1 and zero-shot columns are given in Table 3. Degradation factor = zero-shot RMSE / Exp. 1 RMSE; wet area is the mean SFINCS land-inundation area over the 10 test runs of that event.

Table 7. Full test-set metrics, Exp. 1 (same event)

Event	Mode	Active cells				Land cells			
		MAE	RMSE	CSI 0.1	CSI 0.3	MAE	RMSE	CSI 0.1	CSI 0.3
Irma	Single-step	0.0070	0.0214	0.971	0.985	0.0047	0.0141	0.956	0.974
Irma	Rollout	0.1254	0.3697	0.717	0.705	0.1025	0.3364	0.589	0.521
Irma	Persistence	0.0288	0.0963	0.957	0.947	0.0106	0.0404	0.938	0.917
Matthew	Single-step	0.0060	0.0185	0.971	0.986	0.0039	0.0123	0.950	0.971
Matthew	Rollout	0.0890	0.2696	0.731	0.737	0.0682	0.2335	0.578	0.519
Matthew	Persistence	0.0259	0.0926	0.954	0.943	0.0085	0.0340	0.926	0.896
Ian	Single-step	0.0037	0.0185	0.986	0.990	0.0012	0.0074	0.959	0.966
Ian	Rollout	0.0372	0.1613	0.874	0.892	0.0127	0.0735	0.667	0.668
Ian	Persistence	0.0279	0.1065	0.943	0.923	0.0065	0.0386	0.853	0.769
Milton	Single-step	0.0044	0.0156	0.978	0.988	0.0023	0.0091	0.955	0.968
Milton	Rollout	0.0792	0.2835	0.689	0.725	0.0572	0.2564	0.442	0.408
Milton	Persistence	0.0252	0.0946	0.956	0.936	0.0070	0.0373	0.915	0.847
Dorian	Single-step	0.0037	0.0149	0.981	0.988	0.0015	0.0073	0.950	0.963
Dorian	Rollout	0.0400	0.1421	0.826	0.850	0.0167	0.0847	0.583	0.572
Dorian	Persistence	0.0302	0.1182	0.936	0.915	0.0062	0.0375	0.858	0.767
Beryl	Single-step	0.0050	0.0182	0.969	0.983	0.0028	0.0105	0.937	0.955
Beryl	Rollout	0.0669	0.2211	0.712	0.738	0.0360	0.1489	0.470	0.399
Beryl	Persistence	0.0258	0.1022	0.937	0.923	0.0051	0.0270	0.896	0.851
Mean	Single-step	0.0050	0.0178	0.976	0.987	0.0027	0.0101	0.951	0.966
Mean	Rollout	0.0730	0.2412	0.758	0.775	0.0489	0.1889	0.555	0.514
Mean	Persistence	0.0273	0.1017	0.947	0.931	0.0073	0.0358	0.898	0.841

MAE / RMSE in m; CSI = TP / (TP + FP + FN), wet = $h > \text{threshold}$ (0.1 / 0.3 m). There are 24,467 active cells and 21,933 land cells. Each event row pools its 10 test runs $\times$ 168 time steps; the Mean rows are the arithmetic means of the six event rows.

Table 8. Full test-set metrics, Exp. 2 (cross-event, 6-fold leave-one-out); each row is the held-out event of that fold

Event	Mode	Active cells				Land cells			
		MAE	RMSE	CSI 0.1	CSI 0.3	MAE	RMSE	CSI 0.1	CSI 0.3
Irma	Single-step	0.0079	0.0242	0.969	0.984	0.0051	0.0155	0.952	0.970
Irma	Rollout	0.1349	0.4395	0.690	0.708	0.1038	0.4043	0.541	0.509
Irma	Persistence	0.0288	0.0963	0.957	0.947	0.0106	0.0404	0.938	0.917
Matthew	Single-step	0.0071	0.0218	0.967	0.984	0.0043	0.0137	0.945	0.967
Matthew	Rollout	0.0700	0.1999	0.726	0.771	0.0495	0.1634	0.568	0.559
Matthew	Persistence	0.0259	0.0926	0.954	0.943	0.0085	0.0340	0.926	0.896
Ian	Single-step	0.0044	0.0209	0.985	0.988	0.0015	0.0084	0.957	0.961
Ian	Rollout	0.0624	0.2233	0.766	0.796	0.0313	0.1532	0.490	0.467
Ian	Persistence	0.0279	0.1065	0.943	0.923	0.0065	0.0386	0.853	0.769
Milton	Single-step	0.0051	0.0178	0.976	0.986	0.0026	0.0100	0.950	0.962
Milton	Rollout	0.0911	0.3149	0.692	0.736	0.0534	0.2449	0.436	0.414
Milton	Persistence	0.0252	0.0946	0.956	0.936	0.0070	0.0373	0.915	0.847
Dorian	Single-step	0.0044	0.0174	0.980	0.985	0.0016	0.0087	0.948	0.954
Dorian	Rollout	0.0820	0.2916	0.785	0.779	0.0334	0.1734	0.517	0.448
Dorian	Persistence	0.0302	0.1182	0.936	0.915	0.0062	0.0375	0.858	0.767
Beryl	Single-step	0.0060	0.0205	0.964	0.979	0.0030	0.0113	0.930	0.945
Beryl	Rollout	0.0837	0.2893	0.672	0.717	0.0392	0.1755	0.369	0.320
Beryl	Persistence	0.0258	0.1022	0.937	0.923	0.0051	0.0270	0.896	0.851
Mean	Single-step	0.0058	0.0204	0.974	0.984	0.0030	0.0113	0.947	0.960
Mean	Rollout	0.0874	0.2931	0.722	0.751	0.0518	0.2191	0.487	0.453
Mean	Persistence	0.0273	0.1017	0.947	0.931	0.0073	0.0358	0.898	0.841

MAE / RMSE in m; CSI = TP / (TP + FP + FN), wet = $h > \text{threshold}$ (0.1 / 0.3 m). There are 24,467 active cells and 21,933 land cells. Each event row pools its 10 test runs $\times$ 168 time steps; the Mean rows are the arithmetic means of the six event rows.

Table 9. U-Net architecture (7,765,921 parameters)

Layer	In ch.	Out ch.	Kernel
Input	—	13	—
Encoder block 1 (2 conv) + max pool	13	32	3x3 / 2x2
Encoder block 2 (2 conv) + max pool	32	64	3x3 / 2x2
Encoder block 3 (2 conv) + max pool	64	128	3x3 / 2x2
Encoder block 4 (2 conv) + max pool	128	256	3x3 / 2x2
Bottleneck (2 conv)	256	512	3x3
Up-conv 4 + skip concat + Decoder block 4 (2 conv)	512	256	2x2 / 3x3
Up-conv 3 + skip concat + Decoder block 3 (2 conv)	256	128	2x2 / 3x3
Up-conv 2 + skip concat + Decoder block 2 (2 conv)	128	64	2x2 / 3x3
Up-conv 1 + skip concat + Decoder block 1 (2 conv)	64	32	2x2 / 3x3
Output conv	32	1	1x1

Input channels $C = 13$: bed elevation 1 + water-depth look-back 3 + rainfall look-back 3 + seaward water level 3 + river discharge 3. Water level and discharge are scalars, each broadcast to a constant map at the corresponding time.

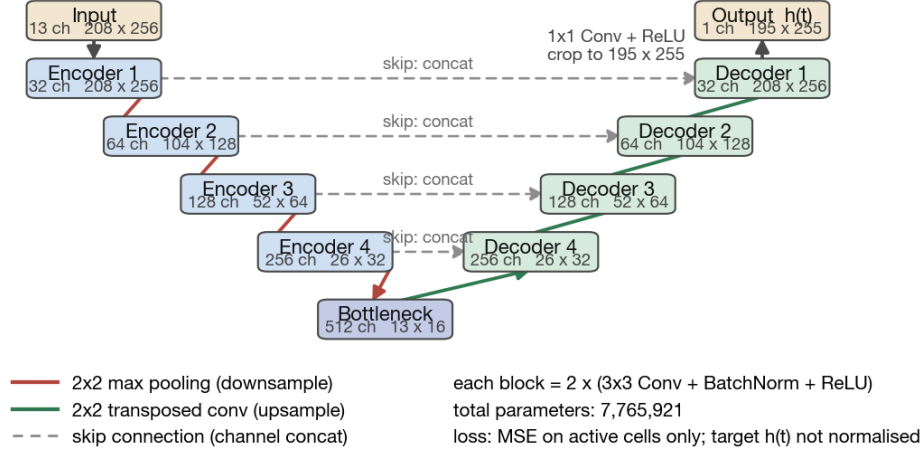


Figure 12. U-Net architecture. Input 13 channels $\times$ 208 $\times$ 256; encoder 32 / 64 / 128 / 256 with a 512 bottleneck (13 $\times$ 16); a symmetric decoder, followed by a 1×1 convolution + ReLU and cropping back to $1 \times 195 \times 255$; 7,765,921 parameters. The channel counts and spatial sizes shown in the figure were read out after instantiating the code.

Table 10. U-Net training setup and hyper-parameters

Item	Setting
Parameters	7,765,921
Hardware	Exp. 1 Apple M4 (MPS); Exp. 2 / 3 NVIDIA A100
Inference time per time step	11.4 ms (measured on the local Mac MPS; ≈ 4.2 ms scaled to the A100)
<i>Pre-training</i>	
Optimiser	Adam
Learning rate	$1e-3$
LR schedule	None; fixed throughout
Loss	MSE masked to active cells
batch size	8
Max epochs	50
Early stopping	Validation loss flat for 10 epochs
Training sampling	One time step every 7 h; 24 per run
Input normalization	Training-set z-score, per channel group
<i>Fine-tuning (Exp. 3)</i>	
Full fine-tuning / frozen encoder / from scratch	LR $1e-4$ / $1e-4$ / $1e-3$; epochs 20 / 20 / 50